

LA TONTOUTA, New Caledonia

WMO: 915900

Lat: 22.017S

Lon: 166.222E

Elev: 36

StdP: 14.68

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	54.1	55.7	48.3	50.2	64.2	49.8	53.1	63.7	22.1	71.8	20.3	71.6	2.1	140	0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.9	93.0	76.2	91.2	75.8	89.6	75.0	79.6	88.1	78.6	86.6	77.6	85.2	11.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.0	83.6	76.5	138.2	82.6	75.4	133.4	81.3	43.3	88.4	42.1	86.9	41.2	84.9	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	20.0	17.5	50.2	96.7	2.3	1.8	48.5	98.0	47.2	99.1	45.9	100.1	44.2	101.5		
(5)			48.9	81.7	1.9	1.3	47.5	82.6	46.4	83.4	45.3	84.1	43.9	85.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.9	79.7	80.1	78.9	75.9	72.1	69.2	67.0	67.1	69.6	72.8	75.6	78.6		
	DBStd	5.59	2.88	2.90	2.80	2.96	3.20	3.12	2.89	3.14	3.28	3.37	3.17	3.07		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	27	0	0	0	0	0	2	10	13	2	0	0	0		
	CDD50	8706	920	842	896	776	686	577	528	530	589	708	768	886		
	CDD65	3259	455	422	431	326	221	129	74	77	142	243	318	421		
	CDH74	25225	4264	3932	3543	2139	1071	488	269	344	859	1790	2626	3900		
	CDH80	8326	1660	1505	1236	578	171	53	17	28	139	494	878	1567		
(14) Wind	WSAvg	6.3	7.2	6.5	6.7	6.1	5.3	5.1	5.0	5.4	6.0	7.2	7.5	7.4		
(15) Precipitation	PrecAvg	37.4	4.8	5.6	4.8	2.9	2.7	3.8	2.5	2.2	1.3	1.8	2.4	2.7		
	PrecMax	74.5	14.9	12.4	13.3	7.5	10.2	9.2	5.7	9.9	6.2	8.4	13.5	10.3		
	PrecMin	16.5	0.6	0.6	0.1	0.1	0.2	0.8	0.3	0.5	0.1	0.0	0.0	0.1		
	PrecStd	11.7	3.7	3.6	3.0	2.3	2.4	2.0	1.4	1.7	1.3	2.1	2.6	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.9	94.7	93.3	90.3	87.0	84.5	82.3	82.9	86.8	90.0	93.0	95.0		
		MCWB	77.0	78.7	77.8	76.4	73.5	71.9	69.6	69.9	71.4	73.0	73.1	76.0		
	2%	DB	91.7	91.8	90.1	87.8	84.1	81.0	79.0	80.3	83.7	87.6	89.7	92.1		
		MCWB	76.1	77.5	77.2	75.3	72.2	69.8	67.3	68.0	69.7	71.8	72.6	74.7		
	5%	DB	89.7	89.8	88.2	85.7	82.0	78.9	77.0	77.4	80.9	85.1	87.5	89.9		
		MCWB	75.1	76.7	76.4	74.3	71.1	68.6	66.2	65.9	68.0	70.2	71.7	73.7		
	10%	DB	87.9	87.9	86.3	83.7	79.5	76.8	75.0	75.5	78.9	82.6	84.6	87.8		
		MCWB	74.7	76.0	75.6	73.4	69.6	67.2	64.9	64.8	66.5	69.1	70.3	73.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.4	81.2	80.7	79.6	76.3	74.2	71.8	72.1	73.8	75.1	76.3	78.7		
		MCDB	89.1	90.4	89.3	86.7	82.7	80.5	78.5	78.5	82.8	86.5	85.5	89.0		
	2%	WB	78.8	79.7	79.1	77.3	74.4	72.0	69.7	70.1	71.7	73.2	74.7	77.2		
		MCDB	87.0	88.5	87.0	84.5	80.1	77.8	75.5	76.6	79.9	83.5	84.9	87.5		
	5%	WB	77.6	78.6	78.1	75.9	72.8	70.5	68.1	68.5	70.0	71.8	73.4	76.0		
		MCDB	85.3	86.6	85.4	82.6	78.4	76.1	73.7	74.6	77.6	81.9	83.3	85.8		
	10%	WB	76.4	77.5	77.1	74.7	71.6	69.1	66.8	66.9	68.4	70.3	72.1	74.9		
		MCDB	83.8	84.6	83.9	80.7	77.3	74.4	72.6	72.7	75.9	79.7	81.6	84.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	15.9	15.3	15.5	15.8	15.9	16.7	17.6	19.4	19.8	19.5	18.9		
		MCDBR	19.5	18.8	18.2	18.5	18.2	18.6	19.4	20.4	22.1	22.8	22.6	22.2		
	5% WB	MCWBR	7.8	7.7	8.0	8.4	8.4	9.1	9.8	10.2	10.4	9.7	9.1	8.6		
		MCWBR	16.2	16.3	16.0	15.9	15.1	15.7	16.3	16.9	19.7	20.7	19.6	18.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.389	0.385	0.385	0.369	0.352	0.344	0.326	0.332	0.352	0.366	0.371	0.378		
		taud	2.548	2.578	2.580	2.610	2.617	2.628	2.657	2.631	2.539	2.522	2.531	2.555		
	Ebn at Noon	Ebn at Noon	303	300	292	284	277	273	283	292	298	303	306	306		
		Edh at Noon	35	33	32	29	26	25	25	28	33	35	35	35		
(44) All-Sky Solar Radiation	RadAvg	2093	1969	1688	1437	1187	1074	1179	1414	1794	2022	2122	2178			
	RadStd	161	146	126	121	86	72	78	99	97	96	143	151			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	-1.10	N/A	N/A	-1.08	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page